Abdelrahman Ehssan

APPLIED AI ENGINEER

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PROFILE

Applied AI engineer building agentic workflows, retrieval-augmented systems, and ML-enabled applications. Experience across LLM orchestration, API integrations, stateful AI workflows, and Python backend engineering.

TECHNICAL SKILLS

AI & ML: LLM agents, RAG, embeddings, HNSW, vector retrieval, LangGraph, LangChain, PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, YOLO

Backend & data: Python, FastAPI, SQL, PostgreSQL/pgvector, Supabase, Pandas, NumPy, Redis, SQLGlot

Infrastructure: Docker, AWS (EC2, S3, RDS), CI/CD, Git/GitHub, Celery, MinIO

EXPERIENCE

Applied AI Engineer | Cycls | Remote (Riyadh, Saudi Arabia) | Aug 2026 – Present

- Engineer LLM-powered agent workflows and backend integrations that connect models with tools and real-world business operations.
- Build stateful discovery and execution flows with structured context, approval boundaries, provider integrations, and reliability-focused validation.

Applied AI Development Intern | Information Technology Institute (ITI) | Jul – Aug 2026

- Developed computer-vision workflows using CNNs, ResNet-18 transfer learning, OpenCV, and YOLO object detection.
- Compared TF-IDF, spaCy, LSTM/GRU and fine-tuned DistilBERT NLP classifiers; examined grounding failures in LLM workflows.

Artificial Intelligence Intern | Technology Innovation & Entrepreneurship Center (TIEC) | Jan – Feb 2026

- Benchmarked six supervised algorithms on 5+ datasets; built reusable preprocessing pipelines that reduced preprocessing time by approximately 40%.

SELECTED PROJECTS

Multi-Tenant Text-to-SQL & Document Chat Platform | ITI Capstone | Aug 2026

- Built a FastAPI/LangGraph platform translating natural-language requests into SQL, with permission-aware schema exposure and SQLGlot validation.
- Implemented multi-format document RAG with pgvector retrieval, reranking, and evidence-backed responses across PDF, DOCX, XLSX, CSV and TXT sources.

LearNova — AI Learning Platform | Team Graduation Project | Jun 2026

- Built AI/backend components of an adaptive learning platform: 4,542 curriculum chunks, Supabase/pgvector retrieval, Gemini embeddings and HNSW indexing; reported Precision@3 of 90.0%.
- Implemented dropout-risk forecasting (reported MAE ~1.82 days) with drift detection, analytics, and AI-generated performance briefings.

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science (SIM Program) | Alexandria University | Graduated Jun 2026

Machine Learning Specialization (DeepLearning.AI / Stanford Online) | Huawei HCCDA-AI | Google AI Essentials